

Diabetes Prediction

Group 14

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1. Abstract

This project aims to develop and evaluate machine learning models for predicting diabetes using the Pima Indians Diabetes Dataset. The dataset contains medical measurements such as glucose level, blood pressure, BMI, and age.

Data preprocessing was performed to handle hidden missing values by replacing zero values with NaN and inputting them using median values. The dataset was then split into training and testing sets, and feature scaling was applied using StandardScaler.

Three supervised learning models were trained and compared: Logistic Regression, K-Nearest Neighbors (KNN), and Random Forest. The models were evaluated using performance metrics such as accuracy, precision, recall, F1 score, and ROC-AUC.

The results showed that the Random Forest model achieved the best performance, with the highest accuracy and F1 score. However, the model still produced some false negatives, which is a critical issue in medical applications.

Overall, this study demonstrates that machine learning can be an effective tool for diabetes prediction, although further improvements are needed to enhance model reliability, especially in detecting diabetic patients.

2. Introduction

Diabetes is one of the most common chronic diseases worldwide and represents a major public health challenge. According to the World Health Organization, the number of people diagnosed with diabetes has increased significantly over the past decades. Early detection of diabetes is essential to prevent serious complications such as heart disease, kidney failure, and vision loss.

Healthcare institutions collect large amounts of patient medical data, including blood glucose levels, blood pressure, and body mass index. This data can be analyzed using machine learning techniques to detect patterns and support medical decision-making.

Machine learning has become an important tool in healthcare because it can analyze complex relationships within medical data and help identify patients at risk of developing diseases. By using predictive models, healthcare providers can improve early diagnosis and provide better preventive care.

In this project, we aim to build and evaluate machine learning models to predict diabetes based on medical measurements from the Pima Indians Diabetes Dataset. The performance of different models will be compared to determine which algorithm provides the most reliable predictions.

3. Business Problem

Type 2 diabetes is one of the most common chronic diseases worldwide and represents a major challenge for healthcare systems. The Pima Indian population has one of the highest rates of type 2 diabetes in the world. Several factors such as genetics, lifestyle, and medical conditions contribute to the development of this disease.

Healthcare institutions collect medical measurements such as glucose level, blood pressure, body mass index (BMI), insulin level, and age during routine medical examinations. These measures can provide important information about a patient's risk of developing diabetes.

The main objective of this project is to use machine learning techniques to build a predictive model that can identify whether a patient is likely to have diabetes based on clinical measurements. Early prediction of diabetes can help healthcare professionals provide earlier treatment and preventive care, which may reduce the risk of serious complications in the future.

This problem can be formulated as a supervised classification problem because the dataset contains labeled data indicating whether a patient has diabetes (1) or not (0).

The goal of this project is to build and evaluate predictive models that can assist healthcare professionals in identifying patients at risk of diabetes and support early diagnosis.

4. Dataset Description

The dataset used in this project is the **Pima Indians Diabetes Dataset**, which is widely used for machine learning research in medical diagnosis. The dataset contains medical diagnostic measurements collected from female patients of Pima Indian heritage aged 21 years and older.

The dataset includes **768 samples and 8 input features**, along with one target variable called **Outcome**, which indicates whether the patient has diabetes.

The features included in the dataset are:

- Pregnancies
- Glucose
- Blood Pressure
- Skin Thickness
- Insulin
- Body Mass Index (BMI)
- Diabetes Pedigree Function
- Age

The target variable **Outcome** is binary:

- 0 → Patient does not have diabetes
- 1 → Patient has diabetes

This dataset allows us to analyze relationships between medical measurements and the likelihood of diabetes.

5. Methodology

5.1 Data Exploration

Before training machine learning models, it is important to explore and understand the dataset. Exploratory Data Analysis (EDA) was performed to analyze the structure of the dataset, the distribution of features, and the relationships between variables.

First, the dataset was inspected to understand the number of samples and features. The Pima Indians Diabetes dataset contains **768 patient records and 8 medical features**, along with one target variable called *Outcome*.

Next, the distribution of the target variable was analyzed to observe the number of patients with and without diabetes. This step helps determine whether the dataset is balanced or imbalanced.

Several visualizations were also used to better understand the data. Histograms were used to analyze the distribution of variables such as glucose level, BMI, and age. These plots help identify possible outliers or unusual values in the dataset.

In addition, a **correlation matrix** was generated to examine the relationships between different features. The analysis showed that some features, such as **glucose level**, have a stronger relationship with diabetes compared to other variables.

Exploring the dataset helps identify potential data quality issues and prepares the data for further preprocessing steps.

The dataset was loaded into the Python environment using the Pandas library. Basic information about the dataset was examined using functions such as `df.shape`, `df.info()`, and `df.describe()`.

The dataset contains 768 observations and 9 columns, including 8 input features and one target variable (Outcome). All features are numerical, with no explicit missing values reported in the dataset. However, further analysis revealed that some features contain zero values that may represent missing data.

These initial steps provided an overview of the dataset structure, data types, and statistical distribution, which is essential before proceeding to further analysis and preprocessing.

5.2 Handling Hidden Missing Values

During the exploratory data analysis, it was observed that some features contain values equal to zero. In medical measurements, certain variables such as glucose level, blood pressure, BMI, skin thickness, and insulin level should not normally have zero values.

These zero values likely represent missing or unrecorded measures rather than actual values. Therefore, these values are treated as hidden missing values.

To handle this issue, the zero values in these columns were replaced with missing values (NaN). After that, the missing values were filled using the median value of the respective feature. The median was chosen because it is more robust to outliers compared to the mean.

Handling these missing values is an important preprocessing step because machine learning models require clean and valid data to produce reliable predictions.

After replacing zero values with NaN and imputing missing values using the median, the dataset contained no missing values and was ready for further analysis.

5.3 Data Preprocessing

After handling missing values, the dataset mainly consists of numerical features, but encoding was applied where necessary to ensure compatibility with machine learning models.

The input features (X) and the target variable (y) were then separated. The target variable *Outcome* represents whether a patient has diabetes (1) or not (0).

The dataset was split into training and testing sets using an 80/20 ratio. The training set was used to train the machine learning models, while the testing set was used to evaluate their performance on unseen data.

Since the dataset contains features with different scales (such as glucose level, BMI, and age), feature scaling was applied using `StandardScaler`. The scale was fitted on the training data and then applied to both training and testing sets to avoid data leakage.

This preprocessing step ensures that the models are trained on clean, normalized, and well-prepared data, which improves model performance and generalization ability.

5.4 Models

In this project, three supervised machine learning models were used to solve the diabetes classification problem: Logistic Regression, K-Nearest Neighbors (KNN), and Random Forest.

Logistic Regression was chosen as a baseline model because it is simple and effective for binary classification problems. K-Nearest Neighbors was used as a distance-based model that classifies data points based on their similarity to nearby observations. Random Forest was selected because it is an ensemble method that can capture complex relationships in data and often provides high performance.

All models were trained using the training dataset and later evaluated using the testing dataset.

The models were implemented using default parameters from the scikit-learn library for example, the Random Forest model was trained with 100 trees (`n_estimators=100`), which helps improve prediction stability and performance.

In this project, special attention is given to model performance in identifying diabetic patients, as misclassification can have serious real-world consequences.

6. Evaluation Metrics

To evaluate the performance of the machine learning models, several classification metrics were used. Accuracy measures the proportion of correct predictions among all predictions. Precision measures the proportion of correctly predicted positive cases among all predicted positives. Recall (Sensitivity) measures the proportion of actual diabetic patients correctly identified by the model. The F1-score provides a balance between precision and recall.

In addition, a confusion matrix was used to visualize the number of true positives, true negatives, false positives, and false negatives. ROC-AUC was also considered to evaluate the model's ability to distinguish between diabetic and non-diabetic patients.

In medical screening tasks, recall is particularly important because a false negative means that a patient with diabetes is incorrectly classified as healthy. This could delay diagnosis and treatment. Therefore, identifying diabetic patients correctly is critical in healthcare applications.

7. Results

In this section, the performance of the trained models is presented and compared.

Three models were evaluated: Logistic Regression, K-Nearest Neighbors (KNN), and Random Forest. The evaluation was based on multiple metrics including accuracy and F1 score.

	Model	Accuracy	F1 Score
0	Logistic Regression	0.804688	0.576271
1	KNN	0.757812	0.523077
2	Random Forest	0.828125	0.666667

Table 1. Performance comparison of the three machine learning models (Logistic Regression, KNN, and Random Forest) based on Accuracy and F1 Score.

Table 1 presents the performance of the three models used in this project. Among them, Random Forest achieved the highest accuracy (0.83) and F1 score (0.67), indicating the best overall performance.

Logistic Regression showed relatively strong performance with an accuracy of 0.80 but a lower F1 score compared to Random Forest. KNN had the lowest performance among the three models, with both lower accuracy and F1 score.

These results suggest that the Random Forest model is more effective for this classification task.

Among all models, Random Forest performed the best in terms of both accuracy and F1 score. This indicates that it provides a better balance between correctly identifying diabetic and non-diabetic patients.

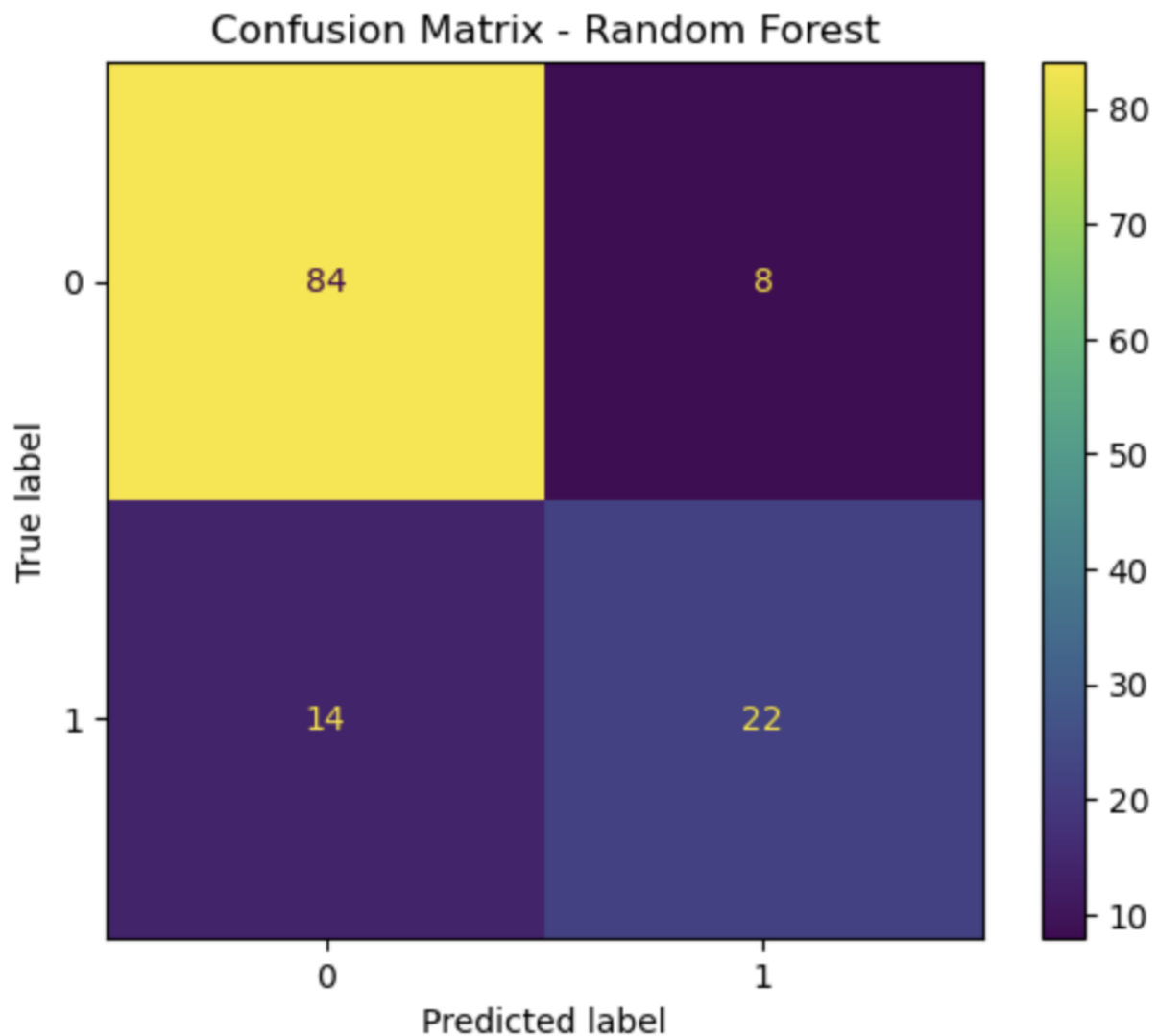


Figure 1. Confusion matrix for the Random Forest model showing the number of correct and incorrect predictions.

Figure 1 shows the confusion matrix for the Random Forest model. The model correctly classified 84 non-diabetic cases (true negatives) and 22 diabetic cases (true positives).

However, it incorrectly classified 8 non-diabetic cases as diabetic (false positives) and failed to identify 14 diabetic cases (false negatives).

This indicates that the model performs well in identifying non-diabetic patients but still has some difficulty detecting all diabetic cases.

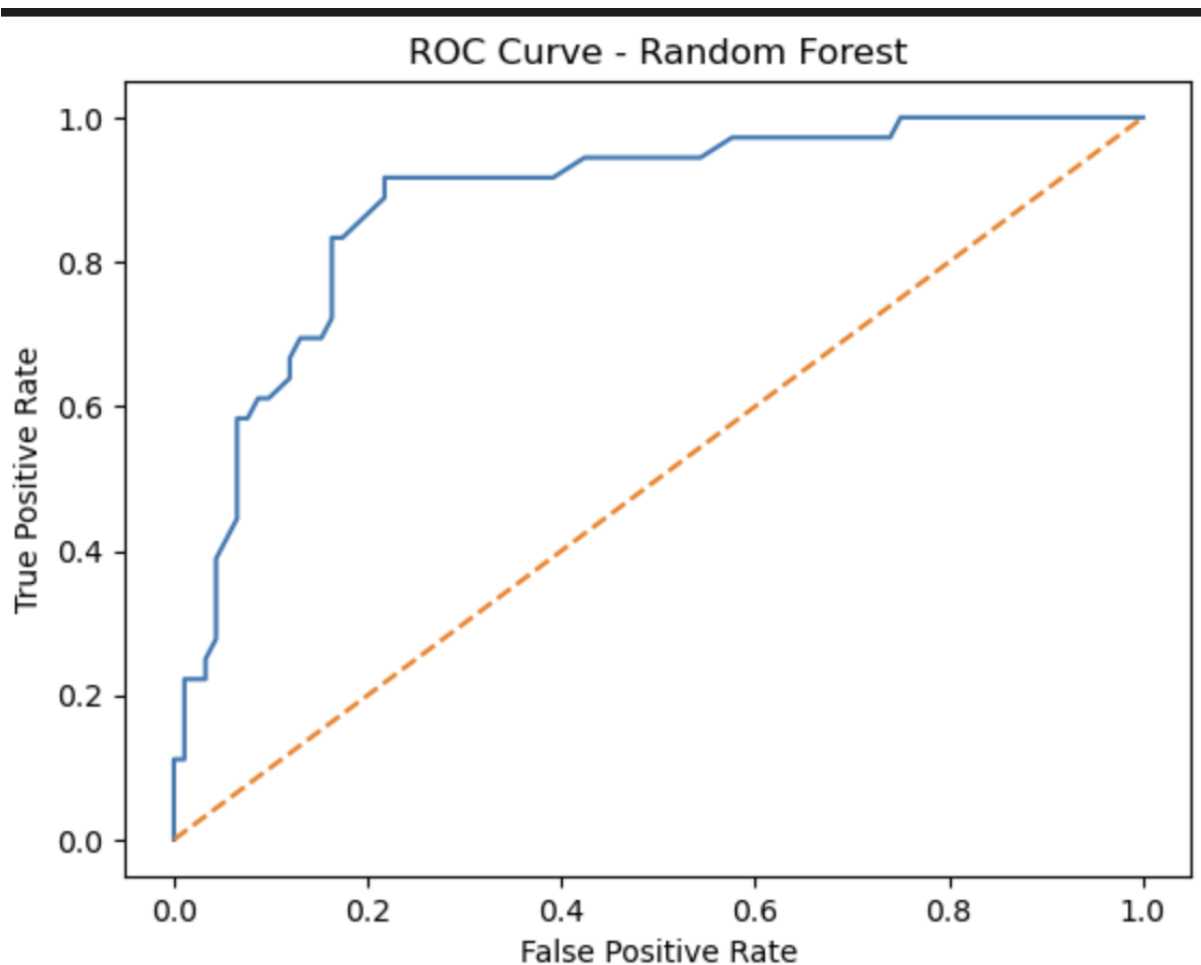


Figure 2. ROC curve for the Random Forest model showing the trade-off between true positive rate and false positive rate.

Figure 2 shows the ROC curve for the Random Forest model. The curve is significantly above the diagonal line, which represents random guessing.

This indicates that the model has good classification performance and is able to distinguish between diabetic and non-diabetic patients effectively.

The ROC AUC score of approximately 0.76 further confirms that the model performs well in separating the two classes.

8. Discussion

The results show that machine learning models can be used to predict diabetes with reasonable performance. Among the three models, Random Forest achieved the best results in terms of accuracy and F1 score, indicating its ability to handle complex patterns in the data.

However, the confusion matrix shows that the model still produces false negatives, meaning that some diabetic patients are incorrectly classified as non-diabetic. This is a serious issue in medical applications because it may delay diagnosis and treatment.

In this context, recall is more important than accuracy, since correctly identifying diabetic patients is critical. A model with higher recall would reduce the risk of missing patients.

Another limitation is that the models were used with default parameters. Better performance could likely be achieved by tuning the models or using more advanced methods.

Overall, while the results are good, there is still room for improvement, especially in reducing false negatives.

9. Conclusion

In this project, machine learning models were used to predict diabetes based on medical data. The dataset was preprocessed and three models were trained: Logistic Regression, KNN, and Random Forest.

The results showed that Random Forest performed the best, achieving the highest accuracy and F1 score. This indicates that it is the most suitable model among those tested.

However, the model still has limitations, especially in identifying all diabetic patients. Future work could focus on improving recall and reducing false negatives by tuning the models or using more advanced techniques.

Overall, this project demonstrates that machine learning can be a useful tool for diabetes prediction.

10. References

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